

Uber Supply-Demand Gap Analysis - Comprehensive Project Report

Project Overview

Objective: Analyze Uber ride request data to identify supply-demand gaps and create an interactive dashboard for key business insights.

Dataset: 6,745 Uber ride requests from July 2016

Tools Used: PostgreSQL, Python, Power BI

1. Data Pipeline Architecture

Pipeline Design

Raw CSV → PostgreSQL (Central Hub) → Python (EDA) + Power BI (Dashboard)

Database Management

- PostgreSQL Setup: Created uber database with uber_requests table
 - Schema Design: 10 columns including derived fields for analysis
 - Data Import: Successfully loaded 6,745 records via pgAdmin command
 - Pipeline Benefit: Live connection enabling real-time dashboard updates
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2. Data Cleaning & Transformation

SQL-Based Cleaning

- Timestamp Standardization: Resolved mixed date-time formats (MM/DD/YYYY vs DD-MM-YYYY)
- Time Extraction: Created request_hour (0-23) for hourly analysis
- Time Slot Categorization: Generated 7 logical time periods (Night to Late Night)
- Business Logic Validation: Confirmed drop_timestamp nulls for failed trips

Data Quality Results

- 100% completeness for operational fields
 - Mixed format resolution: 4,071 DD-MM-YYYY + 2,674 MM/DD/YYYY successfully standardized
 - Derived fields created: request_hour, time_slot for analysis requirements
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3. Exploratory Data Analysis (Python)

Connection & Setup

- PostgreSQL connection established via SQLAlchemy engine
- Data extraction using pandas read_sql for seamless integration
- Live data access ensuring analysis on most current dataset
- Environment setup with essential libraries (pandas, numpy, matplotlib, seaborn)

Key Statistical Findings

- Trip Success Rate: Only 42% completion (2,831/6,745 requests)
- Failure Breakdown: 39% No Cars Available, 19% Cancelled
- Peak Demand: Hour 18 (6 PM) with 510 requests
- Geographic Split: City 52%, Airport 48%

Critical Pattern Discovery

- Airport Crisis: 53% no cars available vs 27% in city
- City Problem: 30% cancellation rate vs 6% at airport
- Evening Supply Gap: 75.8% failure rate during airport rush
- Morning Cancellations: 70.7% failure rate for city early morning

Driver Utilization Analysis

- 300 active drivers handling 4,095 assigned requests
 - 2,650 requests received no driver assignment
 - 31% cancellation rate among assigned rides
 - Night shortage: Only 150 driver assignments vs 1,145 in morning
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4. Dashboard Development (Power BI)

Connection & Setup

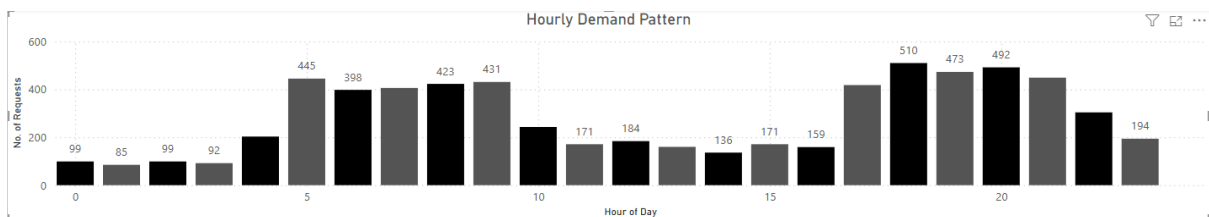
- Live PostgreSQL connection established via Power BI native connector
- Real-time pipeline enabling automatic data refresh
- Professional layout: Single-page dashboard with 5 key visualizations

Visualization Strategy

- Uber Brand Colors: Black/gray color scheme for professional consistency
- Executive Focus: Clean, minimal design emphasizing business insights
- Interactive Elements: Page-level filters for dynamic analysis

Dashboard Visualizations & Insights

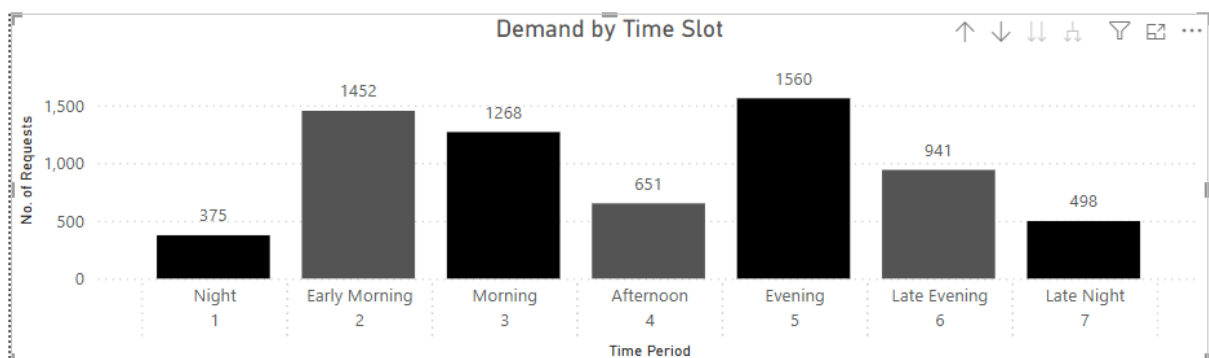
1. Hourly Demand Pattern



Insight: Clear bimodal demand with 6 PM peak (510 requests) and morning surge (5-9 AM).

Business Impact: Evening rush represents maximum system stress requiring capacity focus

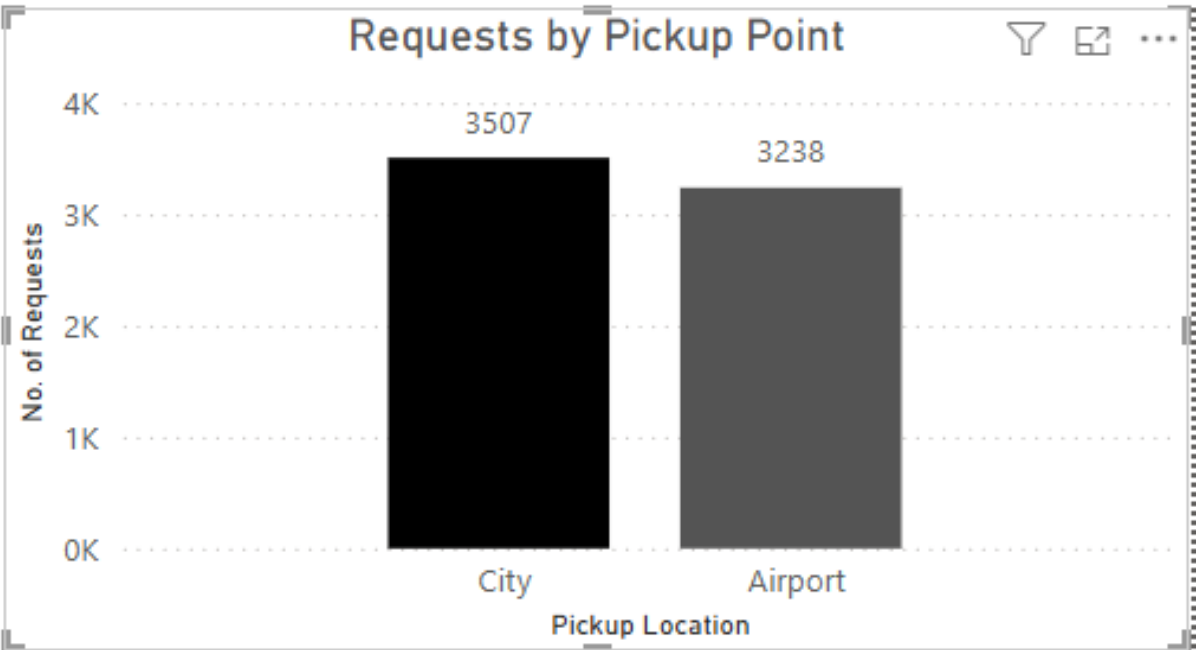
2. Time Slot Distribution



Insight: Evening (1,560) and Early Morning (1,452) account for 45% of total demand.

Business Impact: Resource allocation must prioritize these two critical periods

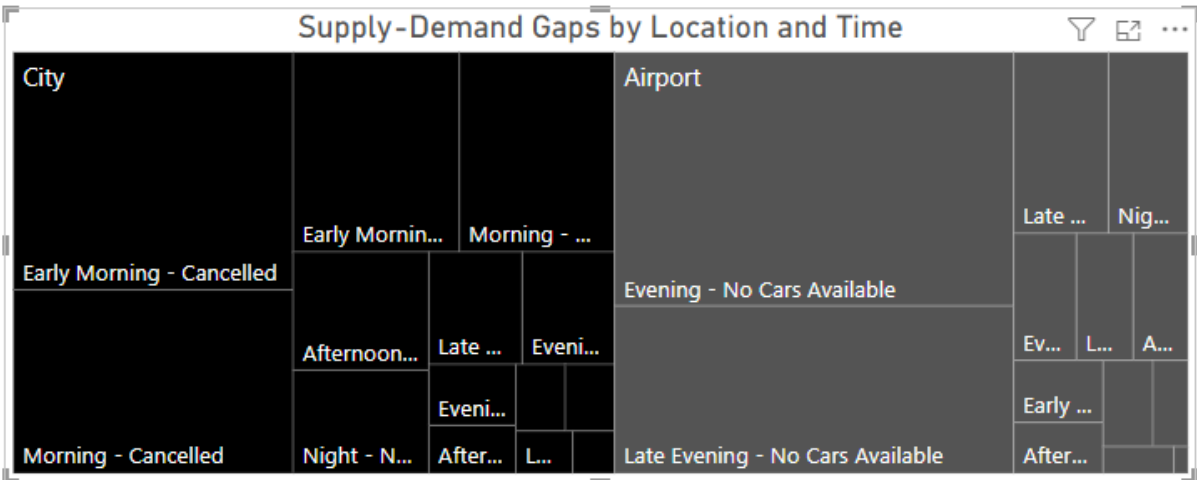
3. Geographic Balance



Insight: Balanced demand distribution - City (3,507) vs Airport (3,238).

Business Impact: Equal operational focus required across both locations

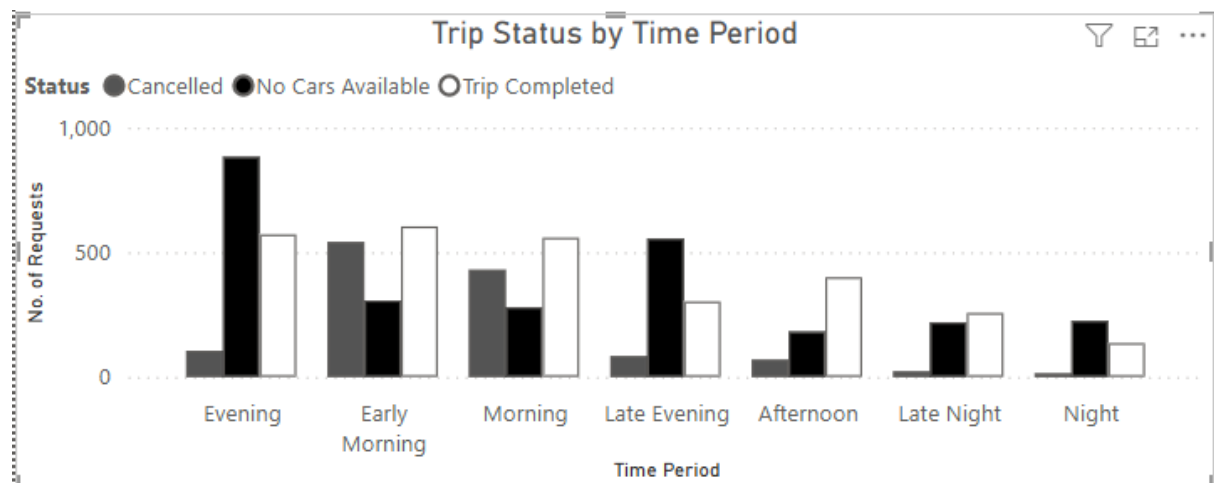
4. Supply-Demand Gap Treemap



Insight: Visual identification of crisis zones - City morning cancellations and Airport evening supply gaps.

Business Impact: Targeted intervention areas clearly defined by location and time

5. Status Breakdown by Time



Insight: Failure rates peak during highest demand periods with different failure types dominating different times.

Business Impact: Time-specific solutions required - supply issues vs cancellation issues

6. Critical Business Findings

Primary Crisis Zones Identified

1. Airport Evening Rush: 81.3% failure rate (586 failed/721 total requests)
2. City Early Morning: 70.7% failure rate (789 failed/1,116 total requests)
3. Night Operations: 60-64% failure rates across both locations

Revenue Impact Assessment

- 58% overall failure rate represents massive opportunity cost
- Peak hour failures occur during highest-value periods
- Geographic inequality in service reliability affects customer experience

Operational Gaps

- Driver shortage: 2,650 requests receive no driver assignment
 - Driver behavior: 31% cancellation rate among assigned rides
 - Capacity misalignment: Supply not matching predictable demand patterns
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7. Technical Implementation Summary

Database Architecture

- PostgreSQL central hub for data storage and management
- Structured schema supporting both operational and analytical needs
- Live connection capability for real-time dashboard updates

Analytics Pipeline

- Python EDA for deep statistical analysis and pattern discovery
- Power BI dashboard for executive-level visual insights

Quality Assurance

- Data validation confirmed business logic and completeness
 - Cross-verification between Python analysis and Power BI visualizations
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Conclusion

This comprehensive analysis successfully identified critical supply-demand gaps in Uber's operations, revealing that airport evening supply shortages and city morning cancellation patterns represent the highest-priority business challenges. The integrated pipeline from PostgreSQL through Python analytics to Power BI dashboard provides a robust foundation for ongoing operational monitoring and strategic decision-making.

Key Success: Transformed raw operational data into actionable business intelligence, enabling targeted interventions that could significantly improve service reliability and revenue performance.
